

Module Code	Examiner	Academic Unit	Tel
CPT208		Computing	



Xi'an Jiaotong-Liverpool University

西交利物浦大學

2nd Semester, 2024/25 Resit Examination

Human-Centric Computing

Time Allowed: 2 Hours

Instructions to Candidates

1. This is an open-book examination with invigilation. You are allowed to take one-page notes with you. Please complete the assessment independently and honestly.
2. Total marks available are **100**.
3. This exam consists of **three questions**. The mark allocated for each question is indicated at the end of the question.
4. Answer **ALL** questions. There is **NO** penalty for providing a wrong answer.
5. Only **English** solutions are accepted. Answer should be written in the answer booklet provided. Clearly indicate the **question numbers** before your solutions. Submit your notes together with your answer booklet. Make sure your writings are clear to read.
6. All materials must be returned to the exam supervisor upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Question 1 Discovering Requirements and Design (40 marks)

You are the lead designer for a new mobile application called *Library+*, which aims to enhance **students' experience with the university library**. The app will allow students to search and reserve books, locate study spaces, receive personalized recommendations, get real-time updates about book availability, and interact with librarians through a live chat feature. Your team is following a human-centred design approach. Answer the following questions based on this scenario.

- (a) Describe the three principles of user-centered process in interaction design. Select one principle and explain its importance in human-centered design, and also specify how to apply it in designing *Library+*. [10]
- (b) Please describe what is persona, and why it is important in human-centered design. Then, please identify and describe one typical persona for *Library+*. [10]
- (c) It is critical to serve both domestic and international students in XJTLU. Please demonstrate what is inclusiveness in human-centered design and why it is important. Then, please explain how *Library+* should be designed to ensure the inclusiveness for all students. [10]
- (d) Define what *interface* means in interaction design. Specifically, for *Library+*, please identify three appropriate interface types and justify why each is suitable, then describe their ideal design implementations and provide concrete examples for each. [10]

Question 2 Prototyping (30 marks)

You are part of a design team developing *FitU*, a mobile application designed to support student **fitness training** on campus. Key features include personalized workout suggestions, gym check-ins, hydration reminders, and sleep tracking. Your team has entered the prototyping phase and is focused on delivering a user-centered experience.

- (a) Please describe what is storyboard and how it could be helpful in the design process of *FitU*. Then, please use this tool to draw four scenarios for *FitU*. [10]
- (b) Define what is visual fidelity in prototyping and discuss two reasons why using high-fidelity visual prototypes might not be appropriate during early design stages. Then, regarding the *FitU* application, please provide and justify one example of a feature that would benefit from high-fidelity visual prototyping, and one example where low-fidelity visual prototyping would be more appropriate. [10]
- (c) Identify and briefly explain the two fundamental aspects of effective screen design. For each aspect you identified, please describe specifically how it should be implemented in the *FitU* mobile application. [10]

Question 3 Evaluation (30 marks)

You are part of a team developing a fitness tracking app called *FitTrack*. The company wants to assess user engagement and satisfaction with the app compared to a competitor app called *HealthMate*. You are asked to **systematically evaluate the two apps using experimental design**.

- (a) How can you apply quantitative and qualitative methods in the evaluation? Name a typical example for each and compare and contrast the differences. Give specific details to explain how the two methods can be used in the given context. [15]
- (b) Your team plans to conduct an experimental study to determine if user engagement and satisfaction with *FitTrack* are significantly higher than with *HealthMate*. Identify the independent variable(s) and dependent variable(s) in this experiment. What are the null hypotheses? Provide a detailed experimental plan that describes your preparations, the experimental settings, what users need to do, how to collect, analyze, and interpret the data. [15]

————— *End of Questions* —————